UNIT 7 · STANDARD 6.EE.7

Solve Multiplication and Division Equations

Lesson 7-3

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Multiplication and division are inverses: $3 \times 7 = 21$, so $21 \div 3 = 7$

Inverse operation

Two math actions that undo each other, like × and ÷.

$3 \times 7 = 21$ means 3 groups of 7 equals 21

Multiply

To add equal groups again and again.

$36 \div 4 = 9$ means 36 split into 4 equal groups gives 9 per group

Divide

To split into equal groups.

$3x = 21$ → divide both sides by 3 → $x = 7$ (x is alone)

Isolate

To get the letter by itself on one side.

Key Ideas & Notes

- Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence.
- She writes the equation $3x = 21$ to find how many pieces are in each box.
- How can she solve for x?
- Solve each equation by showing your work. Use inverse operations to isolate the variable.

Think About It

- What does the 3 represent in the equation?
- What does x represent?
- What operation connects 3 and x?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Park has 3 identical boxes holding 21 pieces of evidence total, written as $3x = 21$. What does each part of the equation mean, and how is solving this different from solving $x + 3 = 21$?

Level 1 support

Start here: In $3x = 21$ the variable is multiplied, so I undo it with division.

The 3 means ____ and x means ____.

El 3 significa ____ y x significa ____.

Solving $3x = 21$ is different from $x + 3 = 21$ because ____.

Resolver $3x = 21$ es diferente de $x + 3 = 21$ porque ____.

Word bank: inverse operation | multiply | divide | isolate

Listen for: *Students read 3x as 3 groups of x, recognize multiplication, and contrast it with addition, knowing they will divide (not subtract) to solve.*

Level 2

Predict: would x in $3x = 21$ be larger or smaller than in $5x = 21$? Justify before solving.

- x would be ____ in $5x = 21$ because ____.
- More boxes means each box has ____.

EXPLORE

How did you undo the multiplication in 3x = 21 (or the division in x / 4 = 9) to isolate the variable?

Level 1 support

Start here: I use the inverse operation: division undoes multiplication.

I divided / multiplied both sides by ___, so the answer is ___.

Dividí / multipliqué ambos lados por ___, así que la respuesta es ___.

The inverse of ___ is ___, so I ___.

El inverso de ___ es ___, así que ___.

Word bank: inverse operation | multiply | divide | isolate

Listen for: *Students divide both sides by 3 to get x = 7 (or multiply by 4 to get x = 36) and name division/multiplication as inverse operations.*

Level 2

Why does dividing only one side of 3x = 21 give the wrong answer? Compare doing it to one side versus both sides.

- Dividing only one side is wrong because ___.
- To keep the equation true I must ___.

CONNECT

When would you solve a multiplication or division equation in real life?

Level 1 support

Start here: These equations find an unknown when items are split into or made from equal groups.

I would use this kind of equation to find ____.

Usaría este tipo de ecuación para hallar ____.

I would ____ both sides because ____.

____ ambos lados porque ____.

Word bank: inverse operation | multiply | divide | isolate

Listen for: *Students name a real context (equal price per item, sharing equally, total from groups) and choose the correct inverse operation to isolate the variable.*

Level 2

Critique: 'To solve x / 5 = 12 you divide both sides by 5.' Explain the error and the correct inverse operation.

- This is wrong because the inverse of division is ____.
- I should ____ both sides by 5, which gives ____.

Guided Examples

Example 1

Solve: $6x = 42$

Solution: Divide both sides by 6: $x = 42 \div 6 = 7$. Check: $6 \times 7 = 42$ ✓

Answer: A. $x = 7$

Example 2

Solve: $n / 3 = 8$

Solution: Multiply both sides by 3: $n = 8 \times 3 = 24$. Check: $24 / 3 = 8$ ✓

Answer: A. $n = 24$

Example 3

Solve: $9m = 63$

Solution: Divide both sides by 9: $m = 63 \div 9 = 7$. Check: $9 \times 7 = 63$ ✓

Answer: A. $m = 7$

Write About the Math

The Writing Revolution

I can explain my steps using the words inverse operation, multiply, divide, and isolate.

1. Kernel Sentence

subject + verb

Model: Inverse operation is two math actions that undo each other, like × and ÷.
Operación inversa es dos operaciones que se deshacen entre sí, como × y ÷.

Write a kernel sentence about inverse operation. Use a subject and a verb.
Escribe una oración base sobre operación inversa. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Inverse operation matters in math
Operación inversa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Inverse operation matters in math because ____.** *Operación inversa importa en matemáticas porque ____.* _______________
but *pero*	**Inverse operation matters in math, but ____.** *Operación inversa importa en matemáticas, pero ____.* _______________
so *entonces*	**Inverse operation matters in math, so ____.** *Operación inversa importa en matemáticas, entonces ____.* _______________

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about inverse operation.
Di un hecho verdadero sobre inverse operation.

Inverse operation ____.

Question
Pregunta

Ask a question about inverse operation.
Haz una pregunta sobre inverse operation.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about inverse operation.
Muestra entusiasmo sobre inverse operation.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with inverse operation.
Dile a un compañero qué hacer con inverse operation.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

The 3 means ____ and x means ____.
El 3 significa ____ y x significa ____.

Solving 3x = 21 is different from x + 3 = 21 because ____.
Resolver 3x = 21 es diferente de x + 3 = 21 porque ____.

I divided / multiplied both sides by ____, so the answer is ____.
Dividí / multipliqué ambos lados por ____, así que la respuesta es ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Solve: y / 7 = 5

A. y = 35

B. y = 12

C. y = 2

D. y = 7

Show your work:

2. A detective splits 54 clues equally among 9 case files. The equation 9c = 54 represents this. What is c?

A. c = 6

B. c = 63

C. c = 45

D. c = 9

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

Sentence starter: Equation: ____. Solving: ____. The answer ____ is reasonable because ____. A division situation with the same answer: ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

Solve: w / 8 = 6

A. w = 48

B. w = 14

C. w = 0.75

D. w = 2

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Try It 1:** A. y = 35 — *Multiply both sides by 7: y = 5 × 7 = 35. Check: 35 / 7 = 5 ✓*
2. **Try It 2:** A. c = 6 — *Divide both sides by 9: c = 54 ÷ 9 = 6 clues per file.*
3. **Exit Ticket:** A. w = 48 — *Multiply both sides by 8: w = 6 × 8 = 48. Check: 48 / 8 = 6 ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Inverse operation is two math actions that undo each other, like × and ÷.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).